

Scrolling Through Change: How Major Societal Events Reshape Instagram Language

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Research Question

Can a language model detect belief shifts in Instagram captioning before versus after major political events, focusing on the 2016 U.S. election, and does the detection of belief shifts differ between influencer accounts and political figure accounts?

Note: We may expand beyond the 2016 United States presidential election to include additional events (e.g., COVID-19) to see how belief shifts vary across contexts.

Hypothesis

Language models can distinguish whether an Instagram caption was written before or after a major societal event (here, the 2016 United States election). We hypothesize that political figure accounts will have more accurate predictions regarding the time frame of the event (pre vs. post) as compared to influencer accounts.

Dataset Documentation

We use a dataset (*captions.csv*) consisting of Instagram posts collected from public accounts of both political figures and influencers. In total, we manually extracted 800 captions from 40 accounts, including 20 politicians and 20 influencers. For each account, we collected 20 captions, evenly split between pre-event and post-event periods, resulting in 10 captions before the election and 10 captions after. We defined time windows around the 2016 U.S. presidential election to capture temporal differences in language surrounding this major societal event. Specifically, we collected pre-election data from November 2014 to November 2015, and post-election data from November 2017 to November 2018 to avoid immediate event-related noise. Each entry in the dataset includes the account username, account type (politician or influencer), the individual's name, the caption text, the date of the post, and a binary label indicating whether the caption was written before or after the election (0 = pre, 1 = post).

To prepare the data for modeling, we ensured the absence of duplicate entries, converted all text to lowercase, and removed URLs and extra spaces. We also conducted several exploratory checks, including verifying the total number of observations, confirming the number of unique accounts, analyzing average caption

length, and checking the distribution of labels and account types. These checks confirmed that the dataset is clean, balanced, and suitable for modeling.

Dataset Limitations & Future Scaling

Note: Our current dataset is sufficient for prototyping but is too small for effectively training deep learning models such as BERT. Thus, we plan to expand our dataset in future work. Specifically, we will use our curated set of accounts as seed users to identify similar influencers and political figures. We will then leverage automated scraping tools such as Instaloader to collect additional captions across a broader set of accounts and time periods. This larger dataset will help us achieve more robust training and improve the model's ability to learn deeper and more generalizable semantic patterns in language related to belief shifts.

Training, Validation, and Test Sets

We split the data at the account level (by username), not the post level, to prevent the model from memorizing individual writing styles. The dataset was divided into 70% training, 10% validation, and 20% test sets, following our TA's recommendation.

Baseline Model: TF-IDF and Logistic Regression

We implemented a baseline model using TF-IDF vectorization (5,000 features) and a logistic regression classifier. We chose this approach because TF-IDF accounts for word importance, while logistic regression is a strong and interpretable baseline. It establishes a reference point for us before applying deep learning methods.

Baseline Metrics

Our baseline model achieved an accuracy of 0.55, with a macro-F1 score of 0.55, recall of 0.55, and precision of 0.55 on the test set. The validation set produced a similar macro-F1 score of 0.56, which indicates consistent performance across splits. These results are slightly above random chance, which means the model captures a weak temporal signal in the captions. The classification report shows relatively balanced performance across classes, with an F1 score of 0.53 for pre-election captions and 0.57 for post-election captions; so, post-event captions are slightly easier to classify. Additionally, the confusion matrix (seen below) shows that pre-event captions are harder to classify (50% accuracy) than post-event captions (60% accuracy), with substantial overlap between classes.

Analysis of Baseline Metrics

While differences in language exist before and after the 2016 U.S. presidential election, they are subtle and not well captured by TF-IDF's word frequency features. This may be due to the limited size of the dataset, the noisy and informal nature of Instagram captions, and the inability of the baseline model to grasp deeper semantic meaning. These findings motivate our use of a more advanced model, BERT, to further understand belief-related language shifts.

Main Takeaway: Our strongest baseline is the TF-IDF and logistic regression model, which achieves a macro-F1 score of 0.55 on the test set, slightly above chance.

Group Comparison: Influencers vs. Politicians

To further evaluate our hypothesis, we analyzed model performance separately for influencer and political accounts. After training the baseline TF-IDF and logistic regression on the full training data, we applied it to the test set and then split the test data by account type. This allowed us to compute performance metrics independently for influencers and politicians using the same trained model for a fair comparison.

We found a distinct difference in performance between the two groups. For influencer accounts, the model achieved an F1 score of 0.49 and an accuracy of 0.49, which is approximately at the chance level. In contrast, for political accounts, the model achieved a significantly higher F1 score of 0.65 and an accuracy of 0.65. These results **support** our hypothesis that belief-related language shifts are more detectable in political accounts than in influencer accounts.

This difference likely arises because political captions tend to be more tied to societal events and contain opinion-driven and ideologically relevant language. Meanwhile, influencer captions are traditionally more focused on lifestyle and personal content, which may not change as strongly in response to major events. As a result, it is logical that temporal signals in influencer captions are weaker and harder for the model to detect, especially with a preliminary model like TF-IDF. With our current beginning dataset, belief-related language shifts are more detectable in political discourse than in general social media content.

Core Model Training: BERT & Loss Curve

We implement a deep learning model using BERT to more closely examine semantic patterns in Instagram captions. Unlike our TF-IDF baseline, which relies on word frequency, BERT uses contextual embeddings to understand meaning, tone, and

relationships between words. We fine-tuned a pretrained BERT model on our dataset to classify captions as pre-event (0) or post-event(1).

Loss Curve

During training, we monitored the training loss to evaluate whether the model was learning effectively. The training loss decreased steadily from 0.68 to 0.53 to 0.36, a clear downward trend. The smooth and consistent decline indicates that the model was successfully learning patterns that help distinguish pre-event and post-event captions.

Note: While we did not explicitly track validation loss during training, we evaluated the model on a held-out validation and test sets. The consistent performance between validation and test results suggests that the model is not severely overfitting despite the relatively small dataset used.

BERT's Metrics

The BERT model achieved an accuracy of 0.66 and a macro-F1 score of 0.66 on the test set, which is an improvement from the baseline model's macro-F1 score of 0.55. This result indicates that incorporating contextual language helps the model better detect belief shifts compared to TF-IDF's frequency-based methods. BERT is able to capture more subtle patterns in tone and meaning.

Additionally, performance differed across account types, with the model achieving an F1 score of 0.62 for influencer accounts and a higher F1 score of 0.72 for political accounts, which supports our hypothesis that belief shifts are more detectable from political figures than influencers.

BERT versus Baseline Performance

Compared to our baseline TF-IDF model, the BERT model shows a clear improvement. The baseline achieved a macro-F1 score of 0.55, while BERT improved this to 0.66, along with an accuracy of 0.66. This suggests that BERT is better able to account for patterns in the data, most likely because it understands context and meaning rather than just word frequency. However, given that the improvement is moderate, belief shifts in Instagram captions are relatively subtle and not always easy to detect.

Analysis of BERT Metrics

The model performs better on pre-event captions (F1 = 0.70) than on post-event captions (F1 = 0.61). This suggests that post-event language may be more varied or less consistent, which makes it harder for the model to classify correctly.

The group-level results further support our hypothesis as the higher performance on political accounts suggests that belief shifts are more clearly reflected in political

language, which tends to be more event-driven and opinion-based. In contrast, influencer captions are often focused on lifestyle content, where changes over time are likely less pronounced and harder for the model to detect.

Main Takeaway: Our core model BERT achieves a macro-F1 score of 0.66 on the test set, improving upon the TF-IDF baseline of 0.55.

Ablation Plan

To truly fathom what the model is learning, we propose an ablation study in which we remove explicit event-related keywords like “election,” “vote,” “Clinton,” and “Trump” from the captions. This experiment will evaluate whether the model is relying on keyword cues associated with the event or whether it is learning true temporal or belief-related language patterns. If performance drops substantially after the removal of highly informative terms, it would suggest that the model is primarily relying on keyword leakage; if performance remains similar, this would indicate that the model is learning deeper linguistic signals that demonstrate shifts in beliefs or discourse over time.

Next steps beyond ablation include conducting a feature analysis. We will examine which words or tokens most strongly drive pre-election versus post-election predictions. In conjunction with one another, the ablation plan and feature help us distinguish between superficial cues and temporal or belief-related cues in the data.

Figures Drafted

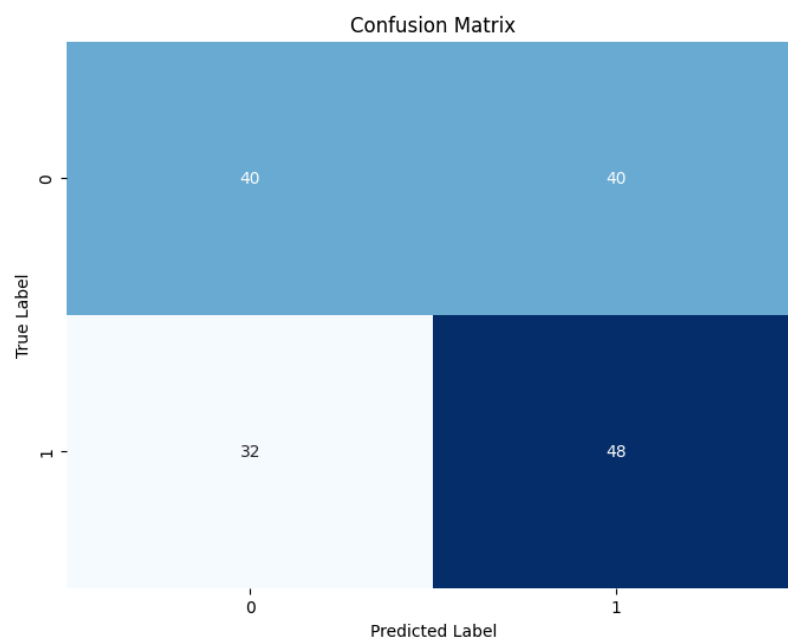


Figure 1. Confusion Matrix for Baseline TF-IDF and Logistic Regression Model.

Confusion matrix for the baseline TF-IDF and logistic regression model, showing substantial overlap between pre-event and post-event classifications.

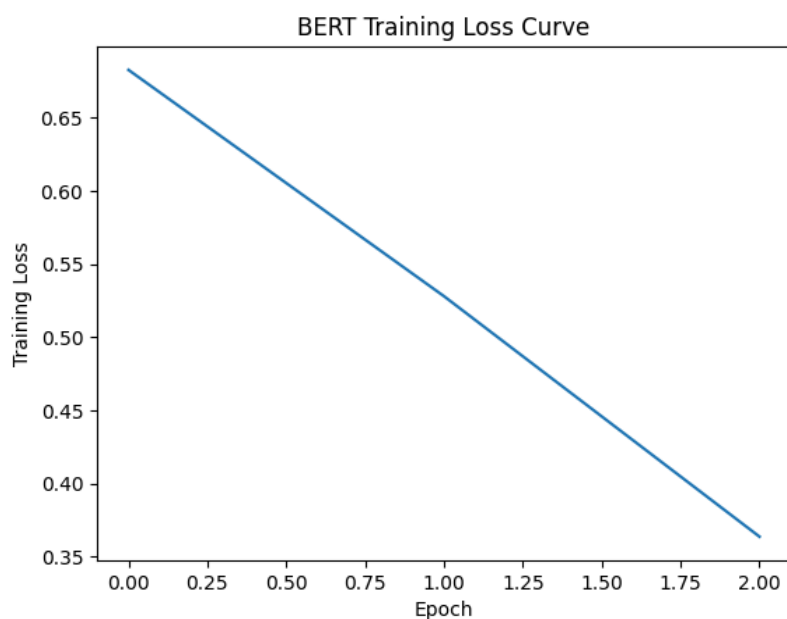


Figure 2. BERT Training Loss Curve. BERT training loss curve, which shows a steady decrease in loss over three epochs (only two graphed).

Next Experiments

- 1) **Dataset Expansion:** We will expand our dataset by using our curated accounts as seed users to identify similar influencers and political figures, and then collect additional captions using Instaloader. This will increase the dataset size, and we expect that it will improve performance by reducing overfitting and more effectively training BERT.
- 2) **Ablation Study:** We will remove explicit event-related keywords from the captions and retrain the model. This tests whether the model depends on keyword leakage or learns temporal belief-related language patterns.
- 3) **Feature Analysis:** We will analyze which words most strongly drive pre-event versus post-event predictions by examining the model's learned feature weights. This will help interpret what the model is learning (i.e., superficial cues vs. language patterns).

Blog Post Background Section

Why is the problem important?

Understanding how beliefs change over time is central to studying cultural and societal evolution. It is difficult to measure belief shifts directly, but language is an observable signal. Language reflects how individuals interpret events, express opinions, frame their opinions, etc. Here, we view analysis of language as a means of approximating changes in beliefs and discourse.

The link between language and belief

Language reflects and shapes how people think. The way individuals describe events reveals how they understand them. Changes in tone, framing, and word choice, among other linguistic features, can indicate shifts in attitudes and perspectives. We are **not** claiming direct belief change, but we treat language as a *proxy* for belief and discourse shifts. If systematic changes in language are observed across many individuals pre-election and post-election, this suggests broader cultural or cognitive shifts.

Capturing cultural change through events

Major societal events (e.g., presidential elections) can reshape public discourse. These events influence how people communicate, how they frame issues, and how they express opinions. By comparing pre-event and post-event language, we can test whether communication patterns change over time. If a model can distinguish between these periods, it suggests that the event produced measurable linguistic changes.

Benefits of Instagram caption analysis

Instagram captions are informal and real-world examples of language with both personal and public expression. They are less filtered than the news, and thus more authentic. They detail everyday reactions rather than curated narratives. Therefore, captions are an excellent means of collecting large-scale data for cultural trend analysis.

Avoidance of the human blind spot

Humans are naturally biased and often unable to see slow, structural changes as they happen. We are using Large Language Models like BERT because they can objectively scan thousands of captions to find patterns that are too subtle or too vast for a person to notice across many users. BERT allows for systematic detection of language changes.

Broader implications

Comparing politicians versus influencers helps identify where strong language shifts occur. Differences in model performance suggest differences in how strongly each group contributes to event-driven discourse.

If models detect consistent language shifts, they may provide early indicators of changes in discourse or sentiment. This could be useful for tracking social trends and studying polarization or public response.

Main Takeaway: If a model can reliably distinguish pre-event from post-event captions, this suggests that major societal events leave a measurable linguistic footprint in social media language, which may reflect broader shifts in discourse and potentially underlying beliefs.

Initial Literature Review

What prior work has been done in this area?

Language shapes thought. The Sapir-Whorf Hypothesis, as we studied in our Cognitive Psychology course, states that language to some degree determines cognition. This foundational theory suggests that the structure of our language affects our worldview. We are applying this to the digital age: if our “Instagram language” changes after an election, does our “Instagram reality” change with it?

Recent computational social science research involving natural language processing and comparative linguistics has further explored how language can be used to study human behavior. In the “From Text to Thought: How Analyzing Language Can Advance Psychological Science” study, researchers show that analyzing text data can reveal

underlying psychological states and cognitive patterns. This supports our approach of using language as a proxy for belief and attitude changes.

Additional studies from 2024, “Highly engaging events reveal semantic and temporal compression in online community discourse,” show that major “shock” events do not just change what we talk about, but they entirely rewire the way we write. These findings suggest that large societal events can reshape how we communicate therefore, reinforcing the importance of our plan to look at the “how” (language) to find the “what” (belief) is on the right track.

Works Cited

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